Lecture 1
Probability and Bayesian Statistics

From Bayesian statistics to probabilistic machine learning

Probabilistic Machine Learning 2027-1

2026-09-30

Section 1

A familiar modeling language

Probabilistic models update based on data

The Beta-Binomial model

Observed

the coin tosses $y_1, \ldots, y_N$

Unknown

the coin bias θ

- A statistical model has data and unknown parameters

The Beta-Binomial model

Observed

the coin tosses $y_1, ..., y_N$

Unknown

the coin bias θ

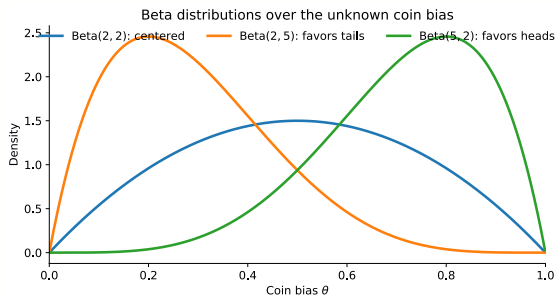
$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$y_i \sim \text{Bernoulli}(\theta), \quad i = 1, ..., N$$

- A statistical model has data and unknown parameters
- How could observed data (y) have been generated.
- What does it tell us about the parameters (θ)

Models are built out of distributions

$$\theta \sim \text{Beta}(\alpha, \beta)$$

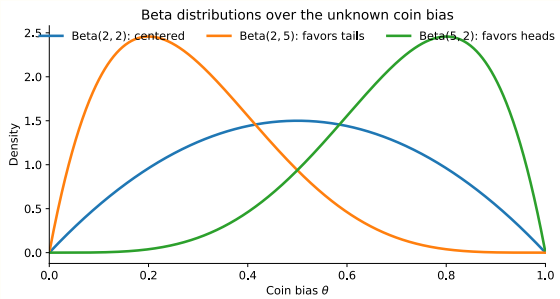


figures/coin_model__beta_distribution.svg

- Beta for continuous variables on $[0, 1]$.

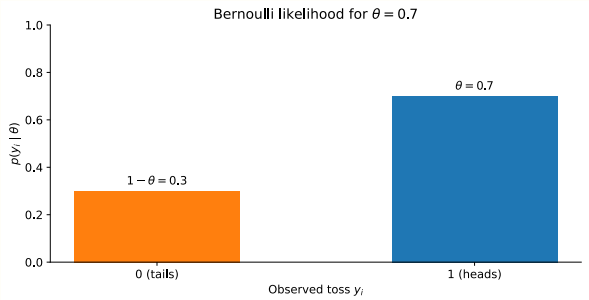
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$$i = 1, ..., N$$

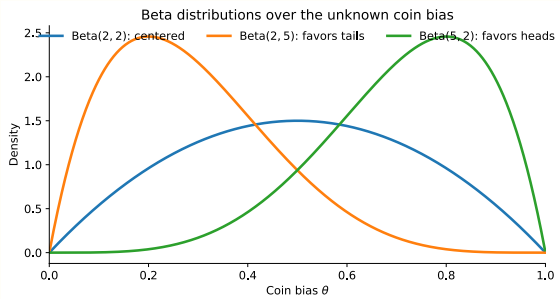


figures/coin_model__bernoulli_likelihood.svg

- Beta for continuous variables on $[0, 1]$.
- Bernoulli four the number of counts in a yes/no process.

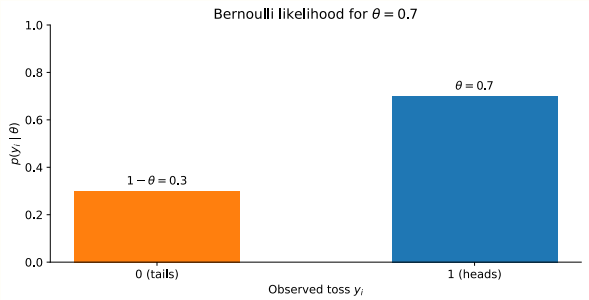
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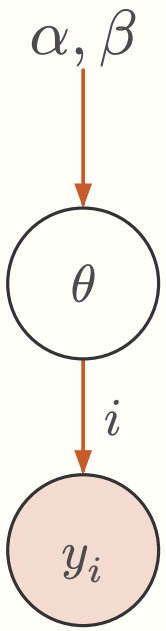
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- Beta for continuous variables on $[0, 1]$.
- Bernoulli four the number of counts in a yes/no process.
- Distributions are parameterized families

Models can be drawn as graphs

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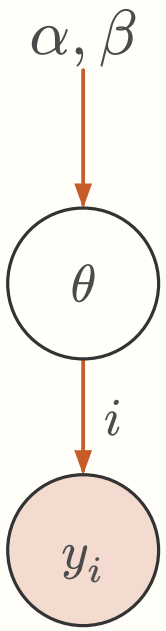


- Nodes represent quantities.
  - Shading marks observed data.

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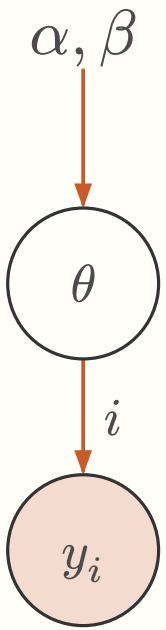


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- Arrows encode conditional dependence.
  - We will not use Kruschke diagrams

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In this course, graphs will allow us to factor complex problems into smaller, more tractable problems

Bayes' rule combines prior and likelihood

Model

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$y_i \sim \text{Bernoulli}(\theta)$$

Bayes' rule combines prior and likelihood

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Bayes' rule

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)}$$

$$\propto p(y \mid \theta)p(\theta)$$

Bayes' rule combines prior and likelihood

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- **Likelihood:** $p(y \mid \theta)$
- **Posterior:** $p(\theta \mid y)$

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- **Prior:** $p(\theta)$
- **Likelihood:** $p(y \mid \theta)$
- **Posterior:** $p(\theta \mid y)$
- We explored the posterior of θ
  - Treating $p(y)$ as a normalizing constant

Conjugacy permits simple calculations

Model

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$$y_i \sim \text{Bernoulli}(\theta)$$

Bayes rule

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Applying Bayes' rule to our model

Conjugacy permits simple calculations

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Applying Bayes' rule to our model

Prior $p(\theta) = \text{Beta}(\alpha, \beta)$

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Applying Bayes' rule to our model

Prior $p(\theta) = \text{Beta}(\alpha, \beta)$

Likelihood $p(y \mid \theta) = \text{Binomial}(N, \theta)$

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Applying Bayes' rule to our model

Prior $p(\theta) = \text{Beta}(\alpha, \beta)$

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Conjugacy

The prior and likelihood are conjugate when the posterior belongs to the same distribution family as the prior.

Conjugacy permits simple calculations

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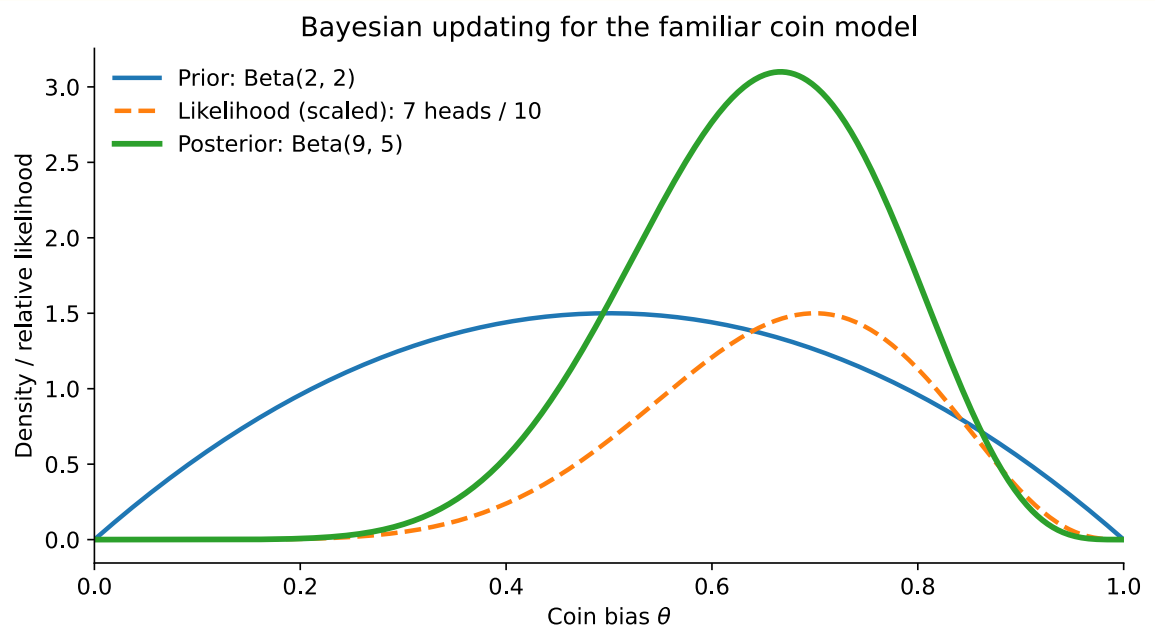
Applying Bayes' rule to our model

Prior	$p(\theta) = \text{Beta}(\alpha, \beta)$
Likelihood	$p(y \mid \theta) = \text{Binomial}(N, \theta)$
Posterior	$p(\theta \mid y) = \text{Beta}(\alpha + y, \beta + N - y)$

Conjugacy

The prior and likelihood are conjugate when the posterior belongs to the same distribution family as the prior.

The posterior is a distribution representing uncertainty



Bayesian updating for the familiar coin model

`figures/coin_model__prior_likelihood_posterior.svg`

- The prior and the posterior are both probability distributions
- Their width expresses the uncertainty
- The uncertainty in the posterior is less than in the prior

Summary: A familiar modeling language

- Probabilistic models link observed data to parameters
- Graphs show model variables and dependencies
- Bayes' rule combines prior and likelihood into a posterior
- Conjugacy makes some updates analytic
- Posterior width represents remaining uncertainty
- Posterior uncertainty narrows with more data

Section 2

From statistics to machine-learning problems

Same probabilistic language, different organizing questions

Statistics focuses on the unknown parameters

$$p(\theta \mid D) = \frac{p(D \mid \theta)p(\theta)}{p(D)}$$

- Get data → build a model → apply Bayes' rule → interpret θ
  - For many scientific questions, that posterior **is** the answer

Statistics focuses on the unknown parameters

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- Get data → build a model → apply Bayes' rule → interpret θ
  - For many scientific questions, that posterior **is** the answer
- Statistics arose from an interest in the underlying parameters in nature

Machine learning grew out of computer science



- A computational task is an **input/output contract**
  - Produce a function f that maps input x to output y

Machine learning grew out of computer science



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  - Multiple implementations may be possible

Machine learning grew out of computer science



- A computational task is an **input/output contract**
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  - Implementation is secondary to the requirement
  - Multiple implementations may be possible
- Machine learning developed in this context

Early machine learning grew out of classification problems

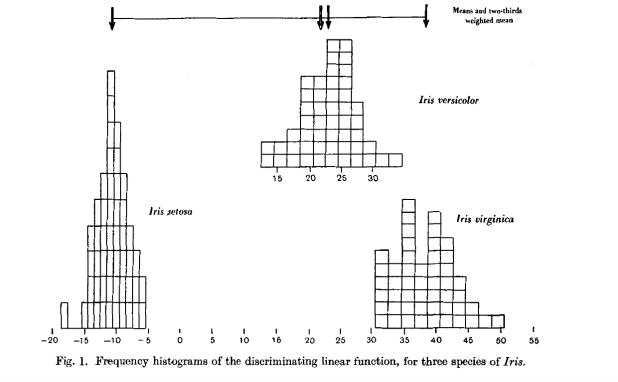


Fig. 1. Frequency histograms of the discriminating linear function, for three species of *Iris*.

Fisher (1936), Fig. 1, The Use of Multiple Measurements in Taxonomic Problems

Every flower started as four numbers. Fisher learned one weighted combination of those four numbers so that species became as separated as possible. The histogram shows the flowers after that learned projection.

Early machine learning grew out of classification problems

Rosenblatt considered the perceptron a model of the vertebrate visual system. He showed how neuron-like units could learn to "recognize" classes of input patterns from experience.

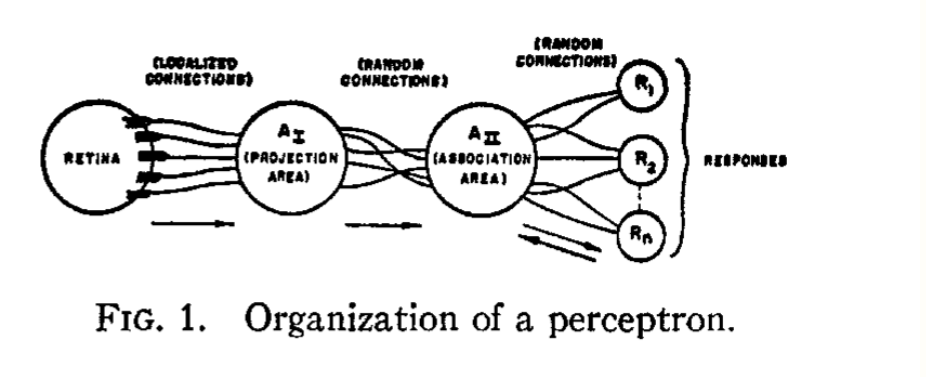


FIG. 1. Organization of a perceptron.

Rosenblatt (1958), Fig. 1, The Perceptron

Early machine learning grew out of classification problems

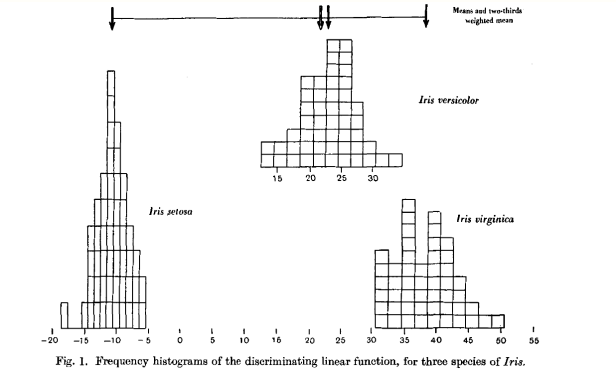


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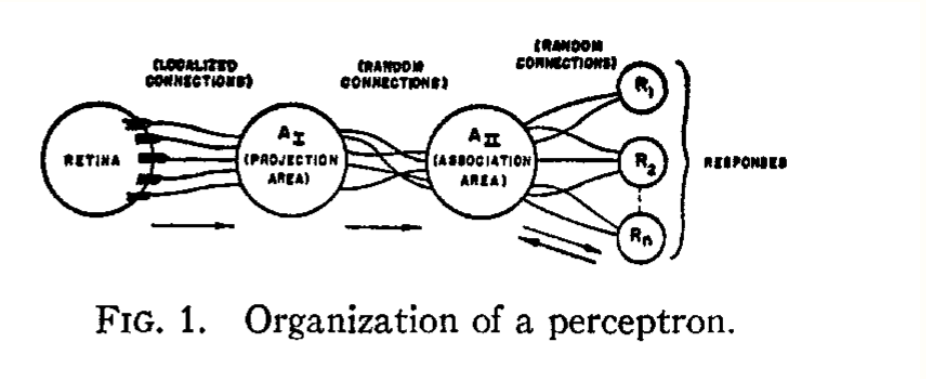
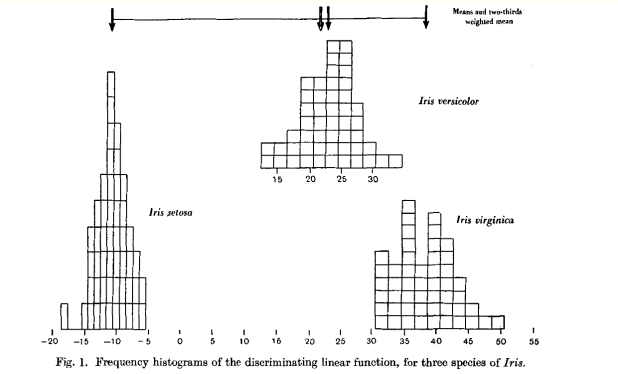


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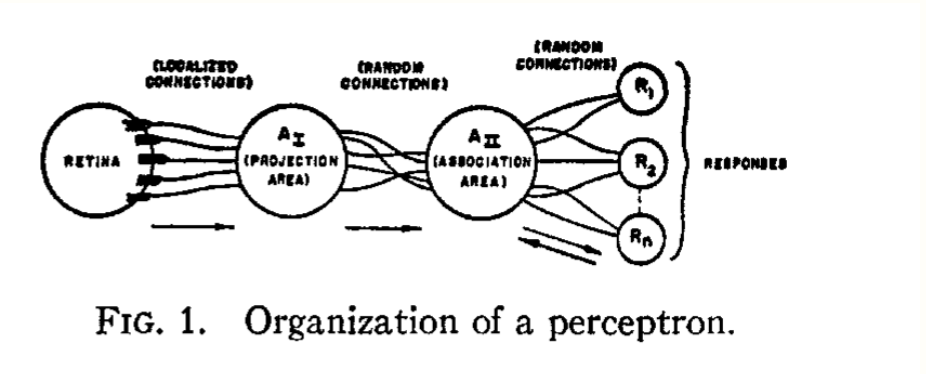
Rosenblatt (1958), Fig. 1, The Perceptron

- Fisher's discriminant and Rosenblatt's perceptron share an interface
  - complex input → class or response

Early machine learning grew out of classification problems



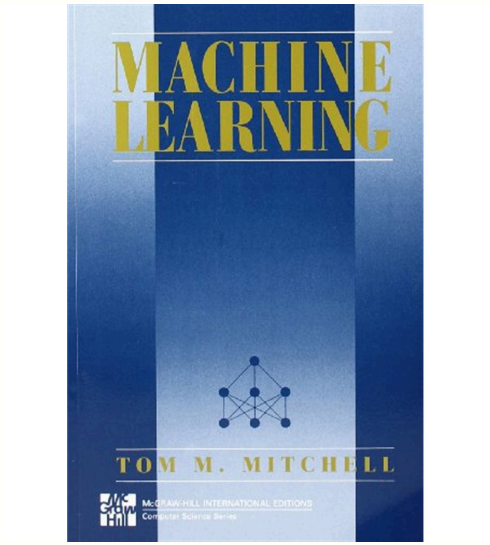
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Rosenblatt (1958), Fig. 1, The Perceptron

- Fisher's discriminant and Rosenblatt's perceptron share an interface
  - complex input → class or response
- Machine learning asks how data can produce the required function f

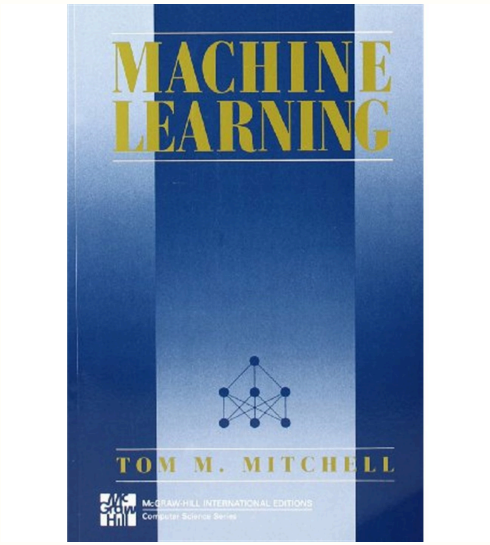
Machine learning defined



Tom M. Mitchell, Machine Learning (1997), cover

A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E. ,

Machine learning defined



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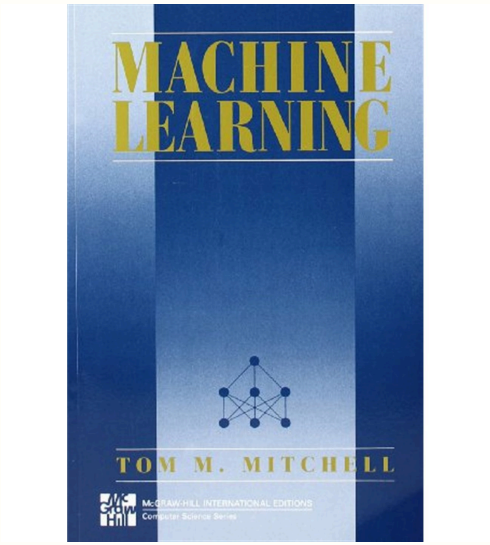
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Training

$$f_T : \mathrm{E} \quad \rightarrow \quad f_A$$

E: Experience

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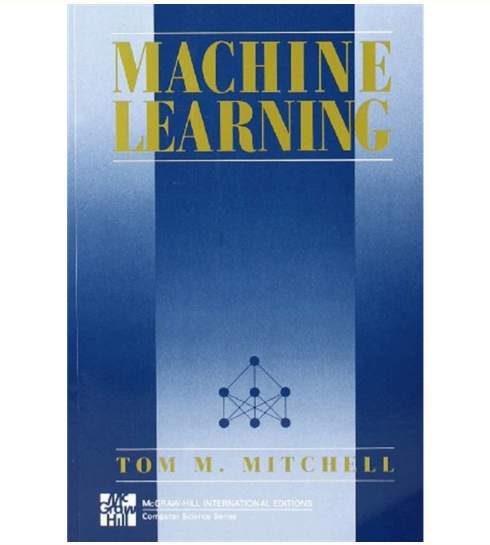
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E: Experience

Task $f_A : \mathrm{In} \rightarrow \mathrm{Out}$

f_A: Task function

Machine learning defined

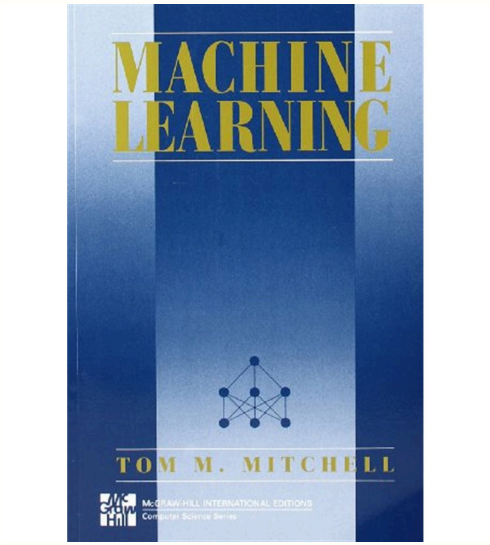


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Training	**Task**	**Evaluation**
$f_T : \mathrm{E} \rightarrow f_A$	$f_A : \mathrm{In} \rightarrow \mathrm{Out}$	$f_P : (\mathrm{In}, \mathrm{Out}) \rightarrow \mathrm{P}$
E: Experience	f_A: Task function	P: Performance

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E: Experience	f_A: Task function	P: Performance

- Machine learning is a mapping from $(E, f_P) \rightarrow f_T \rightarrow f_A$
  - That maximizes f_A performance according to P

Supervised experience supports prediction

Experience: paired examples

$$f_T : \mathrm{E} \quad \to \quad f_A$$

$$f_T : D = \{(x_i, y_i)\}_{i=1}^{N} \quad \to \quad f_D$$

paired examples → learned predictor

Task: prediction

$$f_A : \mathrm{In} \quad \to \quad \mathrm{Out}$$

$$f_D : x_* \quad \to \quad \hat{y}_*$$

new input → predicted output

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- Experience is a training set $D = \{(x_n, y_n)\}$
  - x: called feature, covariate, or predictor
  - y: called label, target, or response
- Task is to produce a predictor f_D that maps new x_* to correct y_*
- Performance is measured by a loss, introduced shortly

Classification: Iris flowers


Setosa


Versicolor


Virginica

Table 1.1: subset of the Iris design matrix

index	sl	sw	pl	pw	label
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
dots	dots	dots	dots	dots	dots
50	7.0	3.2	4.7	1.4	Versicolor
dots	dots	dots	dots	dots	dots
149	5.9	3.0	5.1	1.8	Virginica

sl: sepal length; sw: sepal width; pl: petal length; pw: petal width

Murphy (2025), Book 1, Fig. 1.1 and Table 1.1

- Classification: $y \in \{1, ..., C\}$
  - The classes are unordered and mutually exclusive

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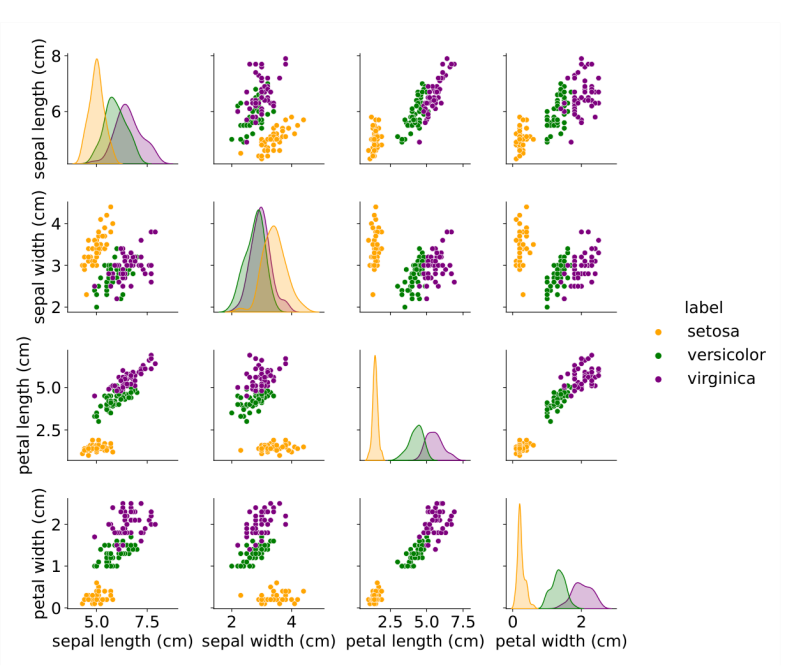
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- Classification: $y \in \{1, ..., C\}$
  - The classes are unordered and mutually exclusive
- For the Iris dataset
  - $y \in \{\text{Setosa}, \text{Versicolor}, \text{Virginica}\}$
  - $x \in \mathbb{R}^4$

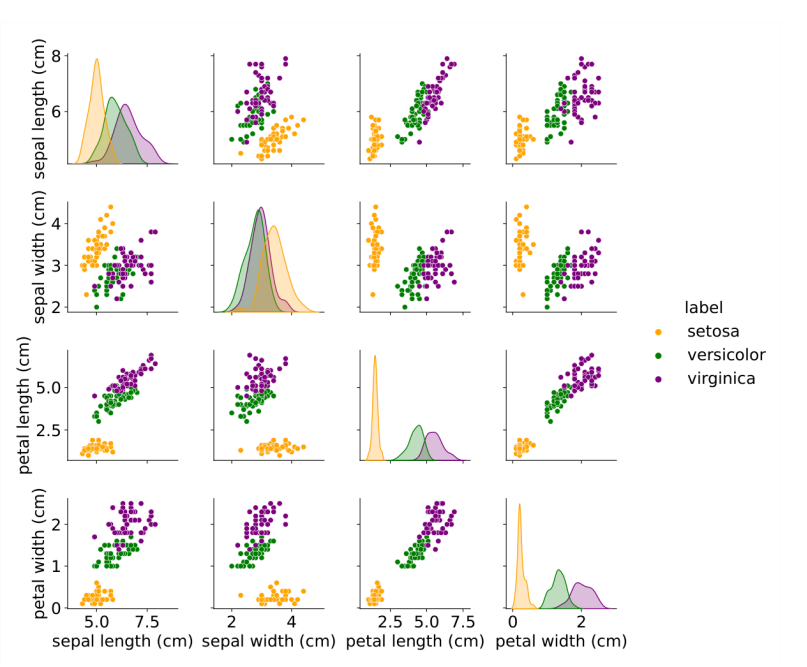
Exploratory data analysis



Murphy (2025), Book 1, Fig. 1.3; generated by iris_plot.ipynb

- Training set
  - 150 labeled examples
  - 50 in each class

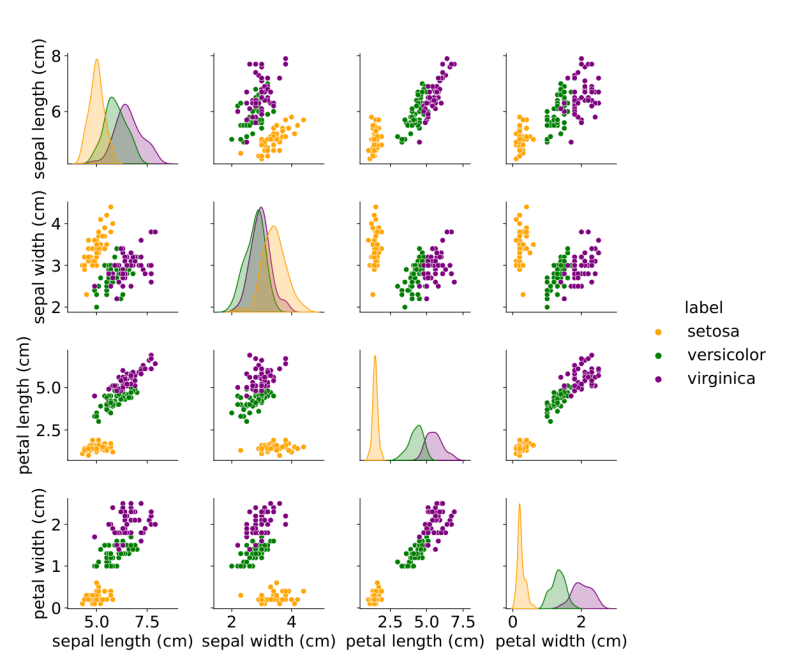
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- Pair plots shows every pair of features
  - Diagonal shows marginal distributions

Exploratory data analysis



Murphy (2025), Book 1, Fig. 1.3; generated by iris_plot.ipynb

- Training set
  - 150 labeled examples
  - 50 in each class
- Pair plots shows every pair of features
  - Diagonal shows marginal distributions
- Always inspect data before modeling
  - Structure
  - Outliers
  - Missing values

Performance: empirical risk minimization

$$L(x, y; \theta) = \ell(y, f(x; \theta))$$

- The loss function: the cost of predicting $f(x; \theta)$ when the true label is y

Performance: empirical risk minimization

$$L(x, y; \theta) = \ell(y, f(x; \theta))$$

$$L(\theta) = \frac{1}{N} \sum_{n=1}^{N} L(x_n, y_n; \theta)$$

- The loss function: the cost of predicting $f(x; \theta)$ when the true label is y
- The empirical risk: loss evaluated on the training set

Performance: empirical risk minimization

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$$\hat{\theta} = \arg\min_{\theta} L(\theta)$$

- The loss function: the cost of predicting $f(x; \theta)$ when the true label is y
- The empirical risk: loss evaluated on the training set
- Empirical risk minimization: minimizing the empirical loss

Performance: empirical risk minimization

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$$\hat{\theta} = \arg\min_{\theta} L(\theta)$$

Given samples $D = \{x^1, ..., x^N\}$ from $p(X)$, where $X \in \mathbb{R}$, the empirical distribution is $\hat{p}_{N(x)} = \frac{1}{N} \sum_{n=1}^{N} \delta_{x^n}(x)$.

- The loss function: the cost of predicting $f(x; \theta)$ when the true label is y
- The empirical risk: loss evaluated on the training set
- Empirical risk minimization: minimizing the empirical loss

A loss function

$$L(x, y; \theta) = \ell(y, f(x; \theta))$$

$$\ell(y, f(x; \theta)) = \ell_{01}(y, \hat{y}) = \mathbb{I}(y \neq \hat{y})$$

- A loss function assigns a cost to a prediction when the true label is known

A loss function

$$L(x, y; \theta) = \ell(y, f(x; \theta))$$

$$\ell(y, f(x; \theta)) = \ell_{01}(y, \hat{y}) = \mathbb{I}(y \neq \hat{y})$$

$$\mathbb{I}(e) = \begin{cases} 1 \text{ if } e \text{ is true} \\ 0 \text{ if } e \text{ is false} \end{cases}$$

- A loss function assigns a cost to a prediction when the true label is known
- The zero-one loss is one for a misclassification and zero for a correct classification
- $\mathbb{I}(e)$ is an indicator function: it evaluates whether condition e holds

A probabilistic perspective

$$p(x \mid y = c, \theta)$$

- The exploratory data analysis suggests:
  - Features in each class have similar distributions
  - Different classes have different distributions

A probabilistic perspective

$$p(x \mid y = c, \theta)$$

$$p(y = c \mid x, \theta) \propto p(x \mid y = c, \theta) p(y = c)$$

- The exploratory data analysis suggests:
  - Features in each class have similar distributions
  - Different classes have different distributions
- Bayes' rule converts those ingredients to posterior class probabilities

A probabilistic perspective

$$p(x \mid y = c, \theta)$$

$$p(y = c \mid x, \theta) \propto p(x \mid y = c, \theta) p(y = c)$$

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} p(y = c \mid x; \theta)$$

- The exploratory data analysis suggests:
  - Features in each class have similar distributions
  - Different classes have different distributions
- Bayes' rule converts those ingredients to posterior class probabilities
- A possible classifier: choose the class with maximum a posteriori probability

Task, experience, and performance

Task

What must the learned system do?

$$x \in \mathbb{R}^4 \to \hat{y} \in \mathcal{Y}$$

four measurements → species

- **Task:** predict the Iris species from four sepal and petal measurements

Task, experience, and performance

Task

What must the learned system do?

$$x \in \mathbb{R}^4 \to \hat{y} \in \mathcal{Y}$$

four measurements → species

Experience

What data, feedback, or interaction is available?

$$D = \{(x_n, y_n)\}_{n=1}^{N}$$

150 labeled flowers

- **Task:** predict the Iris species from four sepal and petal measurements
- **Experience:** 150 labeled flowers, with 50 examples from each species

Task, experience, and performance

Task

What must the learned system do?

$$x \in \mathbb{R}^4 \to \hat{y} \in \mathcal{Y}$$

four measurements → species

Experience

What data, feedback, or interaction is available?

$$D = \{(x_n, y_n)\}_{n=1}^{N}$$

150 labeled flowers

Performance

How is success evaluated?

$$L(\theta) = \frac{1}{N} \sum_{n=1}^{N} \mathbb{I}(y_n \neq f(x_n; \theta))$$

zero-one loss / misclassification rate

- **Task:** predict the Iris species from four sepal and petal measurements
- **Experience:** 150 labeled flowers, with 50 examples from each species
- **Performance:** mean zero-one loss, the training misclassification rate

Summary: From statistics to machine learning

- Statistics and computer science
  - Statistics focused on the values of unknown parameters
  - Computer science develops functions that map inputs to outputs

Summary: From statistics to machine learning

- Statistics and computer science
  - Statistics focused on the values of unknown parameters
  - Computer science develops functions that map inputs to outputs
- Machine learning is a branch of computer science
  - Where the function is learned from experience

Summary: From statistics to machine learning

- Statistics and computer science
  - Statistics focused on the values of unknown parameters
  - Computer science develops functions that map inputs to outputs
- Machine learning is a branch of computer science
  - Where the function is learned from experience
- A learning problem is defined by three ingredients: task, experience, and performance
  - One canonical pairing uses supervised experience $D = \{(x_n, y_n)\}$ for prediction
  - The experience regime and the task are related, but they are not the same taxonomy

Section 3

Classes of machine learning problems

Experience regimes, task classes, and performance criteria

Experience taxonomy

Supervised

$$D = \{(x_n, y_n)\}$$

> Experience describes what the learner gets to observe

Experience taxonomy

Supervised

$$D = \{(x_n, y_n)\}$$

Unsupervised

$$D = \{x_n\}$$

> Experience describes what the learner gets to observe

Experience taxonomy

Supervised

$$D = \{(x_n, y_n)\}$$

Unsupervised

$$D = \{x_n\}$$

Self-supervised

$$D = \{x_n, f(x_n)\} \text{ or } \{f(x_n), x_n\}$$

> Experience describes what the learner gets to observe

Experience taxonomy

Supervised

$$D = \{(x_n, y_n)\}$$

Unsupervised

$$D = \{x_n\}$$

Self-supervised

$$D = \{x_n, f(x_n)\} \text{ or } \{f(x_n), x_n\}$$

Reinforcement / interaction

$$\begin{aligned} \text{decision} &: x(t) \to a(t) \\ \text{action} &: (x(t), a(t)) \to x(t+1) \\ \text{reward} &: x(t+1) \to r(t+1) \end{aligned}$$

> Experience describes what the learner gets to observe

Task taxonomy

Prediction

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

Prediction

Basic task	Map inputs to outputs
Performance	Successful prediction
Example	Classification or regression

Task taxonomy

Prediction

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

Generation

Task: $\{x_n\} \to (\{c\}, p_D(x \mid c)) \to x_* \sim p_D(x \mid c)$

Performance: $f_P(p(x \mid c), x_*)$

Generation

Basic task	Produce new examples
Performance	Similarity to the true distribution
Example	Text or image synthesis

Task taxonomy

Prediction

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

Generation

Task: $\{x_n\} \to (\{c\}, p_D(x \mid c)) \to x_* \sim p_D(x \mid c)$

Performance: $f_P(p(x \mid c), x_*)$

Discovery

Task: $\{x_n\} \to (\{z\}, f_D(x)) \to \hat{z} = f_D(x_*)$

Performance: $f_P(\hat{z}, x_*)$

Discovery

Basic task	Find a representation of the data
Performance	Quality of the discovered representation
Example	Clustering or dimensionality reduction

Task taxonomy

Prediction

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

Generation

Task: $\{x_n\} \to (\{c\}, p_D(x \mid c)) \to x_* \sim p_D(x \mid c)$

Performance: $f_P(p(x \mid c), x_*)$

Discovery

Task: $\{x_n\} \to (\{z\}, f_D(x)) \to \hat{z} = f_D(x_*)$

Performance: $f_P(\hat{z}, x_*)$

Action

Task: $D \to a_* \sim \pi_D(\cdot \mid \{x_{t,*}\}) \to (x_{t+1}, r_{t+1}) = a_t(x_t)$

Performance: $f_P(\{r_t\})$

Action

Basic task	Choose actions to maximize future reward
Performance	Ability to achieve the desired outcome
Example	Reinforcement learning

Task taxonomy

Prediction

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

Generation

Task: $\{x_n\} \to (\{c\}, p_D(x \mid c)) \to x_* \sim p_D(x \mid c)$

Performance: $f_P(p(x \mid c), x_*)$

Discovery

Task: $\{x_n\} \to (\{z\}, f_D(x)) \to \hat{z} = f_D(x_*)$

Performance: $f_P(\hat{z}, x_*)$

Action

Task: $D \to a_* \sim \pi_D(\cdot \mid \{x_{t,*}\}) \to (x_{t+1}, r_{t+1}) = a_t(x_t)$

Performance: $f_P(\{r_t\})$

Prediction	Map inputs to outputs
Generation	Produce new examples
Discovery	Find a representation of the data
Action	Choose actions to achieve desired outcomes

There are many combinations of experience and task

Experience	Prediction	Generation	Discovery	Action
Supervised	Classification	—	—	Imitation learning
Unsupervised	—	Density modeling	Clustering / PCA	—
Self-supervised	Next-token prediction	Language generation	Representation learning	—
Reinforcement / interaction	—	—	—	Policy learning

- Not every type of experience supports every task.
- Common pairings do exist.

Summary: Two taxonomies, three ingredients

- **Learning problem** = task + experience + performance
- **Experience regimes:** supervised, unsupervised, self-supervised, interactive
- **Task classes:** prediction, generation, discovery, action
- Probabilistic modeling provides a unifying language

Section 4

From datasets to learning problems

The same dataset can pose different learning problems

Popular datasets

Iris





AGE
TASKS

Since 1936 – about 90 years
species prediction; clustering; visualization

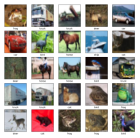
MNIST



AGE
TASKS

Since 1998 – about 28 years
digit classification; representation learning; generation

CIFAR-10



AGE
TASKS

Since 2009 – about 17 years
object classification; robustness; generation

IMDB movie reviews

1. this film was just brilliant casting location scenery story direction everyone's really suited the part they played robert <UNK> is an amazing actor ...
2. big hair big boobs bad music and a giant safety pin these are the words to best describe this terrible movie i love cheesy horror movies and i've seen hundreds...

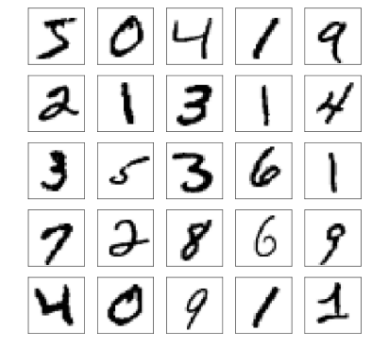
AGE
TASKS

Since 2011 – about 15 years
sentiment classification; text representation

Murphy (2025), Book 1, Figs. 1.1, 1.12a, 1.13b, and Table 1.3

- Machine learning is built around a small number of canonical datasets
  - These four datasets are commonly used
- Each dataset supports multiple tasks
- Datasets support benchmarks that make comparisons easy

MNIST: the same taxonomy, one dataset



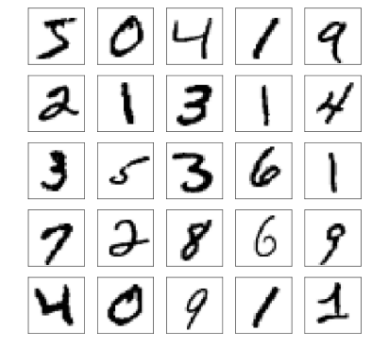
Prediction

$$\hat{y} = f_D(x_*)$$

$$x_n \in \{0, \ldots, 255\}^{28 \times 28}, \quad y_n \in \{0, \ldots, 9\}$$

Murphy (2025), Book 1, Fig. 1.12a

MNIST: the same taxonomy, one dataset



Prediction

$$\hat{y} = f_D(x_*)$$

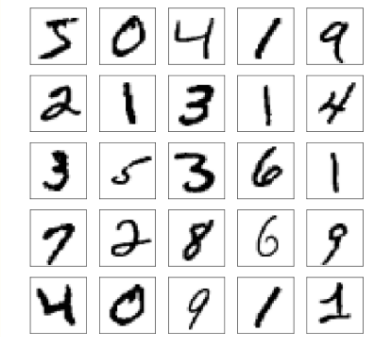
Discovery

$$\hat{z} = f_D(x_*)$$

$$x_n \in \{0, \ldots, 255\}^{28 \times 28}, \quad y_n \in \{0, \ldots, 9\}$$

Murphy (2025), Book 1, Fig. 1.12a

MNIST: the same taxonomy, one dataset



$x_n \in \{0, \ldots, 255\}^{28 \times 28}, \quad y_n \in \{0, \ldots, 9\}$

Prediction

$$\hat{y} = f_D(x_*)$$

Discovery

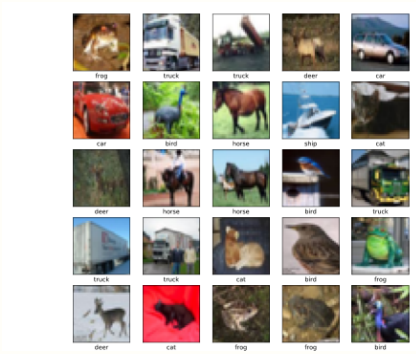
$$\hat{z} = f_D(x_*)$$

Generation

$$x_* \sim p_D(x \mid c)$$

Murphy (2025), Book 1, Fig. 1.12a

CIFAR-10: higher-dimensional inputs, the same questions



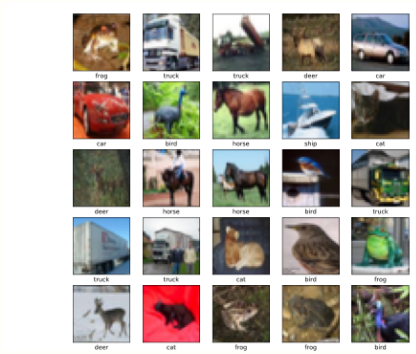
Prediction

$$\hat{y} = f_D(x_*)$$

$$x_n \in \mathbb{R}^{32\times32\times3}, \quad y_n \in \{1, \ldots, 10\}$$

Murphy (2025), Book 1, Fig. 1.13b

CIFAR-10: higher-dimensional inputs, the same questions



Prediction

$$\hat{y} = f_D(x_*)$$

$$L(\theta; D_{\text{train}}) \quad (D_{\text{train}}) \rightarrow L(\theta; D_{\text{test}}) \quad (D_{\text{test}})$$

$$x_n \in \mathbb{R}^{32\times32\times3}, \quad y_n \in \{1, \ldots, 10\}$$

Murphy (2025), Book 1, Fig. 1.13b

IMDB reviews: text as sequence data

1. this film was just brilliant casting location scenery story direction everyone's really suited the part they played robert <UNK> is an amazing actor ...
2. big hair big boobs bad music and a giant safety pin these are the words to best describe this terrible movie i love cheesy horror movies and i've seen hundreds...

Prediction

$$\hat{y} = f_D(x_*)$$

Murphy (2025), Book 1, Table 1.3

IMDB reviews: text as sequence data

1. this film was just brilliant casting location scenery story direction everyone's really suited the part they played robert <UNK> is an amazing actor ...
2. big hair big boobs bad music and a giant safety pin these are the words to best describe this terrible movie i love cheesy horror movies and i've seen hundreds...

Prediction

$$\hat{y} = f_D(x_*)$$

Discovery

$$\hat{z} = f_D(x_*)$$

Murphy (2025), Book 1, Table 1.3

IMDB reviews: text as sequence data

1. this film was just brilliant casting location scenery story direction everyone's really suited the part they played robert <UNK> is an amazing actor ...
2. big hair big boobs bad music and a giant safety pin these are the words to best describe this terrible movie i love cheesy horror movies and i've seen hundreds...

Prediction

$$\hat{y} = f_D(x_*)$$

Discovery

$$\hat{z} = f_D(x_*)$$

Generation

$$w_t \sim p_D(w_t \mid w_{<t})$$

Murphy (2025), Book 1, Table 1.3

Learning problems become probabilistic queries

Prediction

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

Generation

Task: $\{x_n\} \to (\{c\}, p_D(x \mid c)) \to x_* \sim p_D(x \mid c)$

Performance: $f_P(p(x \mid c), x_*)$

Discovery

Task: $\{x_n\} \to (\{z\}, f_D(x)) \to \hat{z} = f_D(x_*)$

Performance: $f_P(\hat{z}, x_*)$

Action

Task: $D \to a_* \sim \pi_D(\cdot \mid \{x_{t,*}\}) \to (x_{t+1}, r_{t+1}) = a_t(x_t)$

Performance: $f_P(\{r_t\})$

- Every task invokes probability models: $p_D(y|x)$, $p_D(x|c)$, $p_D(z|x)$
- Performances are expectations over future cases: $f_P = E[\cdot]$
- Probability models provide a common language for all four task schemas

Summary: From datasets to learning problems

- A dataset is not a task
- A learning problem names variables and a query
  - prediction: summarize $p_D(y_* \mid x_*)$
  - discovery: infer $p_D(z_* \mid x_*)$
  - generation: sample from $p_D(x \mid c)$

Section 5

Probability for machine learning

Expectations, moments, and conditioning

Large learning problems still ask familiar probability questions

Observed
x_{obs}: features, measurements, context

Latent
z: state, representation, missing value

Target
y_*: prediction, sample, decision

- The variables may be high-dimensional, latent, or only partly observed

Large learning problems still ask familiar probability questions

Observed
x_{obs}: features, measurements, context

Latent
z: state, representation, missing value

Target
y_*: prediction, sample, decision

Joint distribution: $p_D(x_{\text{obs}}, z, y_*)$

- The variables may be high-dimensional, latent, or only partly observed
- The task determines which quantity we need from the joint distribution

Large learning problems still ask familiar probability questions

Observed
x_{obs}: features, measurements, context

Latent
z: state, representation, missing value

Target
y_*: prediction, sample, decision



Joint distribution: $p_D(x_{\text{obs}}, z, y_*)$

- The variables may be high-dimensional, latent, or only partly observed
- The task determines which quantity we need from the joint distribution
- The probability language is familiar even when the model is large

A few operations answer many tasks

Joint distribution: $p_D(x_{\text{obs}}, z, y_*)$



Summarize

$\int g(y_*) p_D(y_* \mid x_{\text{obs}}, z)\, \mathrm{d}y_*$

extract a useful quantity

- Prediction and performance often require expectations
- Latent variables are inferred or averaged out
- The same few operations recur across different learning tasks

A few operations answer many tasks

Joint distribution: $p_D(x_{\text{obs}}, z, y_*)$

Summarize

$\int g(y_*) p_D(y_* \mid x_{\text{obs}}, z)\, \mathrm{d}y_*$

extract a useful quantity

Marginalize

$p_D(x_{\text{obs}}, y_*) = \int p_D(x_{\text{obs}}, z, y_*)\, \mathrm{d}z$

average out unwanted variables

- Prediction and performance often require expectations
- Latent variables are inferred or averaged out
- The same few operations recur across different learning tasks

A few operations answer many tasks

Joint distribution: $p_D(x_{\text{obs}}, z, y_*)$

Summarize

$\int g(y_*) p_D(y_* \mid x_{\text{obs}}, z)\, \mathrm{d}y_*$

extract a useful quantity

Marginalize

$p_D(x_{\text{obs}}, y_*) = \int p_D(x_{\text{obs}}, z, y_*)\, \mathrm{d}z$

average out unwanted variables

Condition

$p_D(z \mid x_{\text{obs}}, y_*) = \frac{p_D(x_{\text{obs}}, z, y_*)}{\int p_D(x_{\text{obs}}, z', y_*)\, \mathrm{d}z'}$

update using observations

- Prediction and performance often require expectations
- Latent variables are inferred or averaged out
- The same few operations recur across different learning tasks

Expectation summarizes a distribution

Continuous

$$E[g(X)] = \int g(x)p(x)\,\mathrm{d}x$$

Discrete

$$E[g(X)] = \sum_x g(x)p(x)$$

$$\mu = E[X]$$

Course schematic; Murphy, Book 1 §2.2.5

- Expectation is a probability-weighted average
- Choosing $g(x) = x$ gives the mean

Linearity makes expectations easy to combine

$$E[aX + b] = aE[X] + b$$

$$E\left[\sum_i X_i\right] = \sum_i E[X_i]$$

- Expectations of sums are sums of expectations
- Independence is not required

Variance records spread around the mean

Variance

$$V[X] = E[(X-\mu)^2] = E[X^2] - \mu^2$$

Second raw moment

$$E[X^2] = V[X] + \mu^2$$

Course schematic; Murphy, Book 1 §2.2.5

- Variance measures squared distance from the mean
- The second raw moment contains both spread and location

Moments organize these distribution summaries

Raw moments

$$m_k = E[X^k]$$
$$m_1 = E[X]$$
$$m_2 = E[X^2]$$

Central moments

$$\mu_k = E[(X - E[X])^k]$$
$$\mu_2 = V[X]$$

- A moment is a property of a probability distribution
- Mean and variance are particular moments

Conditioning makes a summary depend on what we know

$Y = y$
observed information

$\rightarrow$

$p(X \mid Y = y)$
$E[X \mid Y = y]$

$$E[X] = E_Y[E[X \mid Y]]$$

- $E[X \mid Y]$ is itself a function of the conditioning variable Y
- Averaging the conditional means recovers the overall mean

Total variance separates within- and between-condition variation

$$V[X] = \underbrace{E_Y[V[X \mid Y]]}_{\text{within}} + \underbrace{V_Y[E[X \mid Y]]}_{\text{between}}$$

$V[X \mid Y]$
conditional spread

+

$E[X \mid Y]$
changing conditional means

Course schematic; Murphy, Book 1 §2.2.5

Summary: Probability for machine learning

- Expectations and moments summarize a probability distribution
- Conditioning makes those summaries depend on observed information
- Total expectation and total variance combine conditional pieces
- These identities are properties of a distribution treated as given

Section 6

From probability to statistics

Estimating unknown quantities from observed data

Probability describes a model; statistics learns from data

Population model

$$Y \sim p(y \mid \theta)$$

unknown θ



Observed sample

$$D = (y_1, \ldots, y_N)$$

What can D tell us about θ ?

- Probability describes possible data given a model
- Statistics uses observed data to learn unknown population quantities

A statistic is a function of the observed data

Data
$$D = (y_1, \ldots, y_N)$$

$\rightarrow$

Statistic
$$T(D)$$

$$\bar{y} = \frac{1}{N}\sum_{n=1}^{N} y_n \quad \text{or} \quad N_1 = \sum_{n=1}^{N} y_n$$

- A statistic is computed from the observed sample
- Its definition does not contain an unknown parameter

An estimator is a statistic aimed at an unknown quantity

$$D \quad \to \quad \hat{\theta}(D)$$

Target
θ

Estimator
$\hat{\theta}(D)$

- The estimator is the rule $D \to \hat{\theta}(D)$
- The estimate is the value produced for the dataset we actually observed
- Different estimation principles can target the same parameter

Sample moments estimate population moments

Population moment

$$m_k = E\left[X^k\right]$$

→

Sample moment

$$\hat{m}_k = \frac{1}{N}\sum_{n=1}^{N} x_n^k$$

$$E[Y] = \theta \quad \text{for Bernoulli data, while} \quad \overline{y} = \frac{N_1}{N}$$

- A population moment is a property of the distribution
- A sample moment is a statistic computed from observed data

Method of moments matches model and sample moments

$$m_{k(\theta)} = \hat{m}_k$$

Bernoulli model

$$E_\theta[Y] = \theta$$

→

Method of moments

$$\hat{\theta}_{\text{MOM}} = \overline{y} = \frac{N_1}{N}$$

Murphy, Book 1 §4.4.1

- Choose parameter values whose model moments equal the empirical moments
- The choice of moments defines an estimation rule

Maximum likelihood uses the sampling model instead

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} p(D \mid \theta)$$

$$p(D \mid \theta) \propto \theta^{N_1}(1-\theta)^{N-N_1}$$

- MLE uses the likelihood rather than selected moments

Maximum likelihood uses the sampling model instead

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} p(D \mid \theta)$$

$$p(D \mid \theta) \propto \theta^{N_1}(1-\theta)^{N-N_1}$$

$$\hat{\theta}_{\mathrm{MLE}} = \frac{N_1}{N} = \hat{\theta}_{\mathrm{MOM}}$$

- MLE uses the likelihood rather than selected moments
- For Bernoulli data the two principles happen to give the same estimate

MAP adds prior information to a point estimate

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid D) = \arg\max_{\theta} p(D \mid \theta) p(\theta)$$

Likelihood
$p(D \mid \theta)$

$\times$

Prior
$p(\theta)$

- MAP is another estimator: it returns the posterior mode

MAP adds prior information to a point estimate

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid D) = \arg\max_{\theta} p(D \mid \theta) p(\theta)$$

Likelihood $p(D \mid \theta)$ × **Prior** $p(\theta)$

$$p(\theta) \propto 1 \quad \rightarrow \quad \hat{\theta}_{\text{MAP}} = \hat{\theta}_{\text{MLE}}$$

- MAP is another estimator: it returns the posterior mode
- A uniform prior reduces MAP to MLE
- The full posterior retains uncertainty that a single MAP estimate discards

Summary: Different estimators answer the same fitting question

Method of moments

$$m_{k(\theta)} = \hat{m}_k$$

Maximum likelihood

$$\max_\theta p(D \mid \theta)$$

MAP

$$\max_\theta p(D \mid \theta)p(\theta)$$

Common goal

$$D \to \hat{\theta}$$

- Statistics provides rules for turning data into estimates of unknown quantities
- MOM, MLE, and MAP use different principles to construct that rule
- A point estimate gives one value, while the posterior retains uncertainty

Section 7

Categorical, multinomial, and softmax

The probability language of multiclass labels

Categorical labels

$$\mathrm{Cat}(y \mid \theta) = \prod_{c=1}^{C} \theta_c^{I(y=c)}$$

$y \in \{1, \ldots, C\}$
one label

→

$\theta \in \Delta^{C-1}$
$0 \leq \theta_c \leq 1$
$\sum_c \theta_c = 1$

One-hot encoding

$y = 2$
class index

$\rightarrow$

$\boldsymbol{y} = (0, 1, 0)^T$
one-hot vector

$$\text{Cat}(y \mid \theta) = \prod_{c=1}^{C} \theta_c^{y_c}$$

Multinomial counts

$$N = (N_1, \ldots, N_C) \quad \sum_c N_c = N \quad \rightarrow \quad \mathrm{Mult}(N \mid N, \theta) \quad \binom{N}{N_1, \ldots, N_C} \prod_{c=1}^{C} \theta_c^{N_c}$$

Course schematic; Book 1 §2.5

Softmax

$a \in \mathbb{R}^C$
logits



Softmax

$a \in \mathbb{R}^C$
logits

→

$\text{softmax}(a)_c = \frac{\exp(a_c)}{\sum_{c'} \exp(a_{c'})}$
normalization

→

Softmax

$a \in \mathbb{R}^C$
logits

→

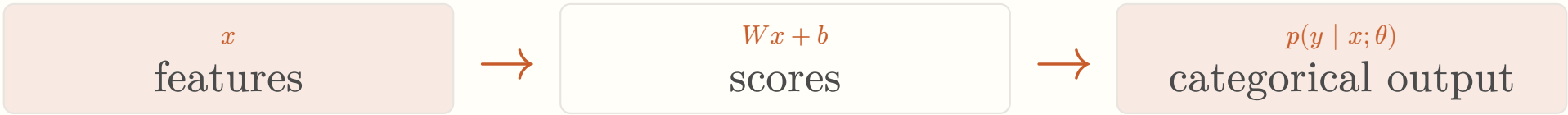
$\text{softmax}(a)_c = \frac{\exp(a_c)}{\sum_{c'} \exp(a_{c'})}$
normalization

→

$\theta \in \Delta^{C-1}$
class probabilities

Multiclass logistic notation

$$p(y \mid x; \theta) = \mathrm{Cat}(y \mid \mathrm{softmax}(Wx + b))$$



Summary: Categorical, multinomial, and softmax

- Categorical: one multiclass label
- One-hot vectors encode class membership
- Multinomial: counts from repeated categorical draws
- Softmax maps logits to the probability simplex
- $p(y \mid x; \theta) = \mathrm{Cat}(y \mid \mathrm{softmax}(Wx + b))$

Section 8

Dirac delta and empirical distribution

Observed data as a distribution of point masses

Point mass

$$\lim_{\sigma \to 0} \mathcal{N}(y \mid \mu, \sigma^2) = \delta(y - \mu)$$

$\mathcal{N}(y \mid \mu, \sigma^2)$
σ^2 tends to 0

→

$\delta(y - \mu)$
all mass at μ

Empirical distribution

$$p_{D(y)} = \frac{1}{N} \sum_{n=1}^{N} \delta(y - y_n)$$

$y_1, \ldots, y_N$
observations

→

p_D
$\frac{1}{N}, \frac{1}{N}, \ldots, \frac{1}{N}$

Course schematic; empirical-distribution definition in the coverage specification

Summary: Empirical distribution

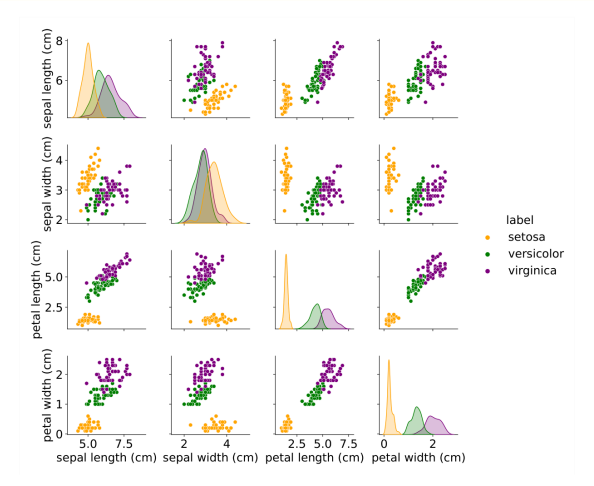
- $\delta(y - \mu)$: a point mass at μ
- $p_{D(y)} = \frac{1}{N} \sum_n \delta(y - y_n)$
- The empirical distribution records observations without smoothing

Section 9

Covariance, correlation, and dependence

Joint variation as geometry

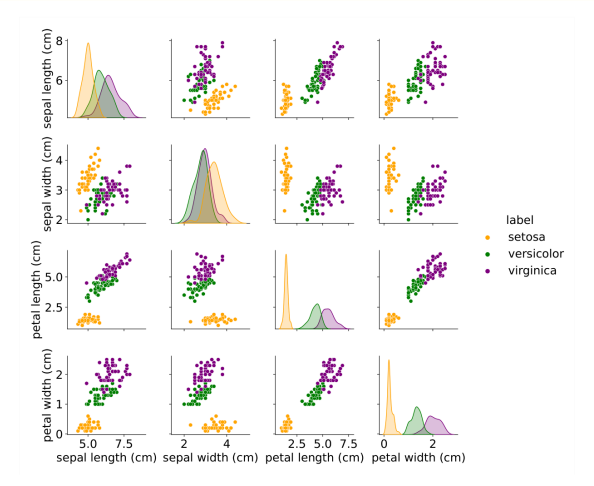
Covariance



$$\mathrm{Cov}[X, Y] = E[(X - E[X])(Y - E[Y])]$$

Murphy (2025), Book 1, Fig. 1.3; generated by iris_plot.ipynb

Covariance



$$\mathrm{Cov}[X, Y] = E[(X - E[X])(Y - E[Y])]$$

> 0	$= 0$	< 0
$+$	0	$-$

Murphy (2025), Book 1, Fig. 1.3; generated by iris_plot.ipynb

Covariance matrix

$$\mathrm{Cov}[\boldsymbol{x}] = E\big[(\boldsymbol{x} - E[\boldsymbol{x}])(\boldsymbol{x} - E[\boldsymbol{x}])^T\big] = \Sigma$$

$$E\big[\boldsymbol{x}\boldsymbol{x}^T\big] = \Sigma + \mu\mu^T$$

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \mathrm{Cov}[X_1, X_2] \\ \mathrm{Cov}[X_2, X_1] & \sigma_2^2 \end{pmatrix}$$

Affine transformations

$\boldsymbol{x}$
random vector

→

$A\boldsymbol{x} + b$
affine map

$$\mathrm{Cov}[A\boldsymbol{x} + b] = A\ \mathrm{Cov}[\boldsymbol{x}]A^T$$

Correlation

$$\rho = \operatorname{corr}[X, Y] = \frac{\operatorname{Cov}[X,Y]}{\sqrt{V[X]V[Y]}}$$

$\rho \approx 1$	$\rho \approx 0$	$\rho \approx -1$
aligned variation	no linear association	opposed variation

Summary: Covariance and dependence

- Covariance: joint variation around the means
- Σ: covariance matrix
- $E[\boldsymbol{x}\boldsymbol{x}^T] = \Sigma + \mu\mu^T$
- $\mathrm{Cov}[A\boldsymbol{x} + b] = A\Sigma A^T$
- Correlation rescales covariance

Section 10

Multivariate Gaussian and conditioning

Geometry, marginalization, and conditioning

Gaussian geometry

$$\mathcal{N}(y \mid \mu, \Sigma) = \frac{1}{(2\pi)^{\frac{D}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(y-\mu)^T \Sigma^{-1} (y-\mu)\right)$$

$\Sigma = \sigma^2 I$

spherical

Gaussian geometry

$$\mathcal{N}(y \mid \mu, \Sigma) = \frac{1}{(2\pi)^{\frac{D}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(y-\mu)^T \Sigma^{-1} (y-\mu)\right)$$

$\Sigma = \sigma^2 I$
spherical

$\Sigma = \mathrm{diag}(\sigma_1^2, \sigma_2^2)$
axis-aligned

Gaussian geometry

$$\mathcal{N}(y \mid \mu, \Sigma) = \frac{1}{(2\pi)^{\frac{D}{2}} \, |\Sigma|^{\frac{1}{2}}} \exp\!\left(-\tfrac{1}{2}(y-\mu)^T \Sigma^{-1} (y-\mu)\right)$$

$\Sigma = \sigma^2 I$	$\Sigma = \mathrm{diag}(\sigma_1^2, \sigma_2^2)$	$\Sigma_{12} \neq 0$
spherical	axis-aligned	tilted

Mahalanobis distance

$$\Delta^2 = (y-\mu)^T \Sigma^{-1} (y-\mu)$$

$y - \mu$
displacement

→

Σ^{-1}
geometry-aware scaling

Typical set

$$\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, I_D), \quad R^2 = \|\boldsymbol{x}\|^2 = \sum_{i=1}^{D} x_i^2$$

$E[R^2] = D$
radius squared

,

$\frac{\mathrm{sd}(R^2)}{E[R^2]} = \sqrt{\frac{2}{D}}$
relative width

Gaussian blocks and marginals

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$p(y_1) = \mathcal{N}(y_1 \mid \mu_1, \Sigma_{11})$
first marginal

$p(y_2) = \mathcal{N}(y_2 \mid \mu_2, \Sigma_{22})$
second marginal

Gaussian conditioning

$$p(y_1 \mid y_2) = \mathcal{N}\left(y_1 \mid \mu_{1|2}, \Sigma_{1|2}\right)$$

Gaussian conditioning

$$p(y_1 \mid y_2) = \mathcal{N}\left(y_1 \mid \mu_{1|2}, \Sigma_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(y_2 - \mu_2)$$

Gaussian conditioning

$$p(y_1 \mid y_2) = \mathcal{N}\left(y_1 \mid \mu_{1|2}, \Sigma_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(y_2 - \mu_2)$$

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Bivariate conditional

$$p(y_1 \mid y_2) = \mathcal{N}\left(y_1 \mid \mu_1 + \rho\frac{\sigma_1}{\sigma_2}(y_2 - \mu_2), \sigma_1^2(1-\rho^2)\right)$$

$y_2 - \mu_2$
observation

→

$\rho\frac{\sigma_1}{\sigma_2}$
mean shift

→

$\sigma_1^2(1-\rho^2)$
remaining variance

Conditioning as an inference machine

$p(y_1, y_2)$
joint Gaussian

→

y_2
observed

→

$p(y_1 \mid y_2)$
Gaussian posterior

$$\Sigma_{1|2} \leq \Sigma_{11}$$

Summary: Multivariate Gaussian and conditioning

- μ: location; Σ: scale, orientation, and dependence
- Mahalanobis distance follows covariance geometry
- High-dimensional Gaussian mass concentrates in a shell
- Marginals retain covariance blocks
- Conditionals are Gaussian with shifted mean and reduced covariance

Section 11

Missing values as Gaussian conditioning

Inference about hidden entries by conditioning

Visible and hidden entries



$$p(y_h \mid y_v, \theta)$$

Summary: Missing values as conditioning

- Partition a vector into visible entries y_v and hidden entries y_h
- Infer hidden entries with $p(y_h \mid y_v, \theta)$
- Gaussian conditioning supplies this distribution analytically

Section 12

Linear Gaussian systems

Latent state, observation, and Gaussian posterior

Linear Gaussian model

$z \sim \mathcal{N}(\mu_z, \Sigma_z)$
latent state

$$p(z, y) = p(z)p(y|z)$$

Linear Gaussian model

$z \sim \mathcal{N}(\mu_z, \Sigma_z)$
latent state



$y \mid z \sim \mathcal{N}(Wz + b, \Sigma_y)$
observation

$$p(z, y) = p(z)p(y|z)$$

Joint Gaussian form

$$\mu = \begin{pmatrix} \mu_z \\ W\mu_z + b \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_z & \Sigma_z W^T \\ W\Sigma_z & \Sigma_y + W\Sigma_z W^T \end{pmatrix}$$

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}(\mu, \Sigma)$$

Posterior state

$$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$$

Posterior state

$$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$$

$$\Lambda_{z|y} = \Sigma_z^{-1} + W^T \Sigma_y^{-1} W$$

Posterior state

$$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$$

$$\Lambda_{z|y} = \Sigma_z^{-1} + W^T \Sigma_y^{-1} W$$

$$\Sigma_{z|y} = \Lambda_{z|y}^{-1}$$

Posterior state

$$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$$

$$\Lambda_{z|y} = \Sigma_z^{-1} + W^T \Sigma_y^{-1} W$$

$$\Sigma_{z|y} = \Lambda_{z|y}^{-1}$$

$$\mu_{z|y} = \Sigma_{z|y}\left(W^T \Sigma_y^{-1}(y - b) + \Sigma_z^{-1}\mu_z\right)$$

Sensor fusion

Prior
$$\mathcal{N}(z \mid \mu_z, \Sigma_z)$$

$\times$

Measurement
$$\mathcal{N}\big(y \mid Wz + b, \Sigma_y\big)$$

$\propto$

Posterior
$$\mathcal{N}\Big(z \mid \mu_{z|y}, \Sigma_{z|y}\Big)$$

$$\Sigma_{z|y}^{-1} = \Sigma_z^{-1} + W^T \Sigma_y^{-1} W$$

Summary: Linear Gaussian systems

- Prior: $p(z) = \mathcal{N}(z \mid \mu_z, \Sigma_z)$
- Observation: $p(y \mid z) = \mathcal{N}\big(y \mid Wz + b, \Sigma_y\big)$
- Joint model: $p(z, y) = p(z)p(y|z)$
- Posterior: $p(z|y)$ is Gaussian
- Precision adds information from the prior and observation

Lecture 1 summary

- ML tasks specify observed quantities, unknown quantities, and required outputs
- Moments and conditional moments describe probabilistic summaries
- Categorical, multinomial, and softmax distributions describe labels
- Covariance gives multivariate geometry
- Gaussian marginalization and conditioning are reusable inference machinery
- Linear Gaussian systems provide the first complete inference machine